KidShield — Product Requirements Document (PRD)

Version: 1.5
Date: March 3, 2026
Author: Product Team
Status: Active — V1 (MVP) Complete, V1.5 In Development

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1. Overview

1.1 Problem Statement

Young children (ages 5–10) are increasingly exposed to addictive apps — social media, short-form video, and games — leading to excessive screen time, reduced physical activity, and attention issues. Existing parental control solutions are either too complex, easily bypassed, or rely on simple PINs that children quickly learn. Worse, most solutions only **block** — they create friction without offering guidance, motivation, or alternatives. Parents themselves often model the very phone addiction they're trying to curb in their children.

1.2 Solution

KidShield is an Android app that combines **face-recognition-based app blocking** with **positive behavioral guidance** for the whole family. When a child attempts to open a blocked app, KidShield displays a friendly, character-driven overlay that not only restricts access but actively **motivates, teaches, and redirects** the child toward healthier activities. For parents, the app provides **usage insights and self-awareness** — surfacing their own screen time alongside their child's, making digital wellbeing a shared family effort rather than a one-sided restriction.

Key differentiators:

- **Not just a wall — a guide.** The blocking overlay features "Buddy," a relatable child mascot character who encourages, teaches, and suggests alternatives
- **Family-first design.** Tracks and surfaces *parent* screen time too — "Kids mirror what they see"
- **Positive reinforcement.** Screen-free streaks, daily report cards, and encouraging messages reward good habits
- **Impossible to bypass.** Face recognition + device admin + accessibility service make circumvention virtually impossible for young children

1.3 Target Audience

Segment	Description
Primary	Parents/guardians of children aged 5–10
Secondary	The children themselves (active participants in building healthier habits)
Tertiary	The family unit as a whole (shared digital wellbeing)

1.4 Platform & Distribution

- **Platform:** Android only (minimum SDK 26 / Android 8.0)
- **Distribution:** Sideloaded APK initially; Google Play Store in a later phase
- **Tech Stack:** Flutter (UI) + Native Android / Kotlin (system services via platform channels)

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2. Goals & Success Metrics

2.1 V1 (MVP) Goals — ☑ COMPLETE

#	Goal	Status
G1	Block access to parent-selected apps via face-verified overlay	☑ Done
G2	Run reliably in background without being killed by the OS	☑ Done
G3	Provide a simple, child-proof setup and management experience	☑ Done
G4	Achieve < 2 second face verification latency on mid-range devices	☑ Done

2.2 V1.5 Goals

#	Goal
G5	Transform the blocking overlay from a cold restriction into a warm, character-driven guidance experience
G6	Track and surface usage statistics (block attempts, screen time) for both child and parent
G7	Introduce positive reinforcement through screen-free streaks and encouraging messages
G8	Provide parents with daily/weekly report cards showing family digital wellness trends
G9	Gently surface parent's own screen time to promote family-wide awareness

2.3 Success Metrics

Metric	Target	Version
Overlay trigger latency (app open → overlay visible)	< 500 ms	V1 ☑
Face recognition accuracy	≥ 97% true-positive, < 1% false-positive	V1 ☑
Background service uptime	≥ 99% (auto-restart on kill)	V1 ☑
Setup completion rate	≥ 90% of onboarding starts	V1 ☑
Average screen-free streak length	> 2 hours	V1.5
Daily blocked app attempt reduction (week-over-week)	≥ 15% decrease	V1.5
Parent engagement with report card	≥ 3x/week views	V1.5
Parent self-awareness feature opt-in rate	≥ 50%	V1.5

3. User Personas

3.1 Parent (Admin User)

- Registers their face (and optionally other guardians' faces) during setup
- Selects which installed apps to block
- Verifies identity via face or fallback PIN to grant temporary app access
- Configures settings such as re-verification interval
- **V1.5:** Reviews daily/weekly report cards showing child's blocked app attempts and screen time trends
- **V1.5:** Optionally sees their own screen time surfaced alongside their child's — a gentle nudge toward modeling good behavior

3.2 Child (Active Participant)

- Uses the device normally for allowed apps
- Encounters a **friendly, character-driven** blocking overlay when opening a restricted app
- **V1.5:** Sees "Buddy" (mascot character) who offers encouragement, fun facts, activity suggestions, and moral lessons
- **V1.5:** Earns recognition for screen-free streaks — building positive habits rather than just hitting walls
- Cannot bypass, uninstall, or disable KidShield

4. Functional Requirements

4.1 Onboarding & Face Registration

ID	Requirement	Priority
F-01	First-launch wizard guides the parent through full setup	P0
F-01a	**Smart onboarding resume**: If the user has already completed certain steps (e.g., face registered, PIN set), the wizard detects this on launch and **skips to the first incomplete step** instead of restarting from the beginning	P0
F-02	Parent registers their face by capturing multiple angles (front, slight left, slight right) using the front camera	P0
F-02a	Front camera images must have **EXIF rotation applied** before processing — `BitmapFactory.decodeByteArray()` does not apply EXIF rotation from JPEG, so rotation must be read via `ExifInterface` and applied via `Matrix` transformation	P0
F-02b	**Skip option**: If faces are already registered, a "Skip" button is available in the AppBar to bypass re-registration	P0
F-03	Face embeddings generated via **MobileFaceNet (mobilefacenet.tflite)** and stored **locally on-device** (encrypted). Model outputs **192-dimensional** embeddings (not 128 as originally documented)	P0
F-04	Support registering **multiple authorized adults** (up to 4 faces)	P0
F-05	Parent sets a **fallback PIN** (6-digit minimum) during setup, used when face detection is unavailable (e.g., poor lighting, broken camera)	P0
F-06	Parent can re-register or delete faces from settings	P1

4.2 Permission Acquisition

ID	Requirement	Priority

ID	Requirement	Priority
F-07	App requests and guides the user to grant the following permissions sequentially with clear explanation screens:	P0
	— **Accessibility Service** (to detect foreground app changes)	P0
	— **Display Over Other Apps** (SYSTEM_ALERT_WINDOW, for blocking overlay)	P0
	— **Device Administrator** (to prevent uninstall)	P0
	— **Camera** (for face recognition)	P0
	— **Usage Stats Access** (PACKAGE_USAGE_STATS, backup detection)	P1
	— **Ignore Battery Optimizations** (to prevent background kill)	P0
	— **Notification Permission** (for persistent foreground service notification)	P0
F-07a	**Battery optimization intent** must use `FLAG_ACTIVITY_NEW_TASK` when launched from a non-Activity context (e.g., platform channel handler), otherwise it crashes on some Android versions	P0
F-07b	**Skip option**: A "Skip for now" button with confirmation dialog allows the parent to proceed without granting all permissions (with a warning that protection won't be fully active)	P0
F-07c	**Restricted settings guidance (Android 13+)**: On sideloaded installs, Android 13+ blocks accessibility service grants due to "restricted settings." The permission screen includes instructions to enable via App Info → "Allow restricted settings"	P0
F-08	If any critical permission is missing, show a persistent banner in-app and block activation of the protection service	P0
F-09	Provide a "permission health check" screen showing the status of each permission with quick-fix buttons	P1

4.3 App Selection (Block List)

ID	Requirement	Priority
F-10	Display a scrollable list of **all installed apps** (excluding system-critical apps like Settings, Phone, Messages) with icons and names	P0
F-11	Parent toggles apps ON/OFF to mark them as blocked	P0
F-12	Provide a **search bar** to filter apps by name	P1
F-13	Show a count of currently blocked apps on the main dashboard	P1
F-14	Persist the block list locally using SharedPreferences or local database	P0
F-15	Newly installed apps default to **unblocked**; parent is notified of new installs to review	P2

4.4 Background Monitoring Service

ID	Requirement	Priority
F-16	A **foreground service** (Kotlin) runs persistently in the background, showing a non-dismissible notification	P0
F-17	The service uses the **Accessibility Service API** to detect when a blocked app's activity moves to the foreground	P0
F-18	On detection of a blocked app, the service **immediately launches the overlay activity** (< 500 ms)	P0
F-18a	**Overlay stacking prevention**: The service maintains an `overlayActive` flag. A new overlay is **never launched** while one is already displayed. This prevents the verification loop where a successful dismiss immediately re-triggers the overlay because the blocked app briefly re-enters foreground	P0
F-18b	**Overlay dismissed callback**: When the overlay is dismissed (via verification, PIN, or navigation), it calls `onOverlayDismissed()` on the accessibility service to clear the `overlayActive` flag and record a grace-period timestamp	P0
F-18c	**Periodic re-check timer**: After the overlay is dismissed and the blocked app is allowed, a `Handler`-based recheck timer periodically verifies the blocked app is still in the foreground. If the user leaves the app, the timer stops. If the re-verification interval expires while the app is still active, the overlay re-appears	P0
F-18d	**Grace period for "every time" mode**: When the re-verification interval is set to 0 (every time), a **3-second grace period** is enforced after each successful verification to prevent immediate re-triggering	P0
F-19	The service **auto-restarts** if killed by the OS (via START_STICKY, WorkManager periodic health check, and a broadcast receiver on BOOT_COMPLETED)	P0
F-20	The service maintains a flag/timestamp for recently verified sessions to avoid re-prompting within the configured interval	P0

4.5 Blocking Overlay & Face Verification

ID	Requirement	Priority
F-21	Overlay is a **full-screen, system-level activity** displayed on top of the blocked app	P0
F-22	Overlay shows: app icon, app name, message ("This app is restricted. Parent verification required."), and a "Verify" button	P0
F-23	Pressing "Verify" activates the front camera and performs **real-time face detection + recognition** using MobileFaceNet	P0
F-24	If the detected face matches any registered parent/guardian (cosine similarity ≥ threshold), the overlay **dismisses** and the blocked app is allowed to continue	P0
F-25	If verification **fails** (face not recognized, no face detected within 10 seconds, or 3 failed attempts), overlay remains and shows a "Use PIN" option	P0
F-26	The overlay **cannot be dismissed** by pressing Back, Home, or Recent Apps — it re-appears immediately	P0
F-26a	Overlay uses `imageProxy.imageInfo.rotationDegrees` from CameraX to correctly orient camera frames for face detection, regardless of device sensor orientation	P0
F-27	If the user navigates away (Home button), the blocked app is sent to background and is not accessible; overlay clears	P0
F-28	After successful verification, the blocked app is **unlocked for a configurable interval** (default: 1800 seconds / 30 minutes) before re-prompting	P0
F-29	PIN fallback: 6+ digit PIN entry screen shown as alternative to face verification	P0

4.6 Settings & Configuration

ID	Requirement	Priority
F-30	Settings screen is **protected** — requires face verification or PIN to access	P0
F-31	**Re-verification interval**: Configurable time (in seconds) after which the overlay re-appears for a previously unlocked app. Options: 0 (every time) / 30s / 60s / 120s / 300s / 900s / 1800s / 3600s, or custom value	P0
F-32	**Manage blocked apps**: Navigate to the app selection screen (F-10)	P0
F-33	**Manage faces**: Add, preview, or delete registered faces	P0
F-34	**Change PIN**: Requires current PIN or face verification first	P1
F-35	**Enable/Disable protection**: Master toggle to pause all blocking (requires verification)	P1
F-36	**Debug/Testing mode**: A hidden developer toggle (accessible via 7-tap on version number) that sets the re-verification interval to custom values (e.g., 10 seconds) for testing	P1

4.7 Mascot Character & Guidance System (V1.5)

ID	Requirement	Priority
F-37	Introduce **"Buddy"** — a relatable human child mascot character that appears on the blocking overlay instead of a cold restriction message	P0
F-38	Buddy is displayed as a **static illustration** with **rotating text speech bubbles** that change each time the overlay appears	P0
F-39	Maintain a **local content library** of 80–120 pre-written messages across the following categories:	P0
	— **Motivational** (e.g., "You're stronger than any screen! Go be awesome!")	P0
	— **Activity Suggestions** (e.g., "How about drawing something cool right now?")	P0
	— **Fun Facts / Did You Know** (e.g., "Did you know octopuses have 3 hearts?")	P0
	— **Moral Micro-Lessons** (e.g., "Being kind to someone today makes the world better.")	P0
	— **Gentle Humor** (e.g., "Even superheroes take breaks from screens!")	P0

ID	Requirement	Priority
F-40	Messages are selected **randomly without immediate repeats** (shuffle-bag algorithm) so the child sees variety	P0
F-41	Messages are **time-of-day aware** where appropriate: morning messages suggest active play, evening messages suggest reading/winding down, weekend messages suggest family activities	P1
F-42	The overlay layout changes from the current clinical design to a **warm, child-friendly design** featuring Buddy prominently, with the verification controls still accessible	P0
F-43	Buddy's illustration has **multiple poses/expressions** (at least 4: waving, thinking, pointing, encouraging) that rotate alongside the messages	P1
F-44	The content library is stored as a **bundled JSON asset** — no internet required, fully offline	P0
F-45	Parents can optionally **disable** the mascot in settings (reverting to the simple overlay)	P2

4.8 Usage Statistics & Tracking (V1.5)

ID	Requirement	Priority
F-46	Track **block events**: each time a blocked app triggers the overlay, record {app_package, timestamp, outcome (face_verified / pin_verified / navigated_away)}	P0
F-47	Track **daily screen time** (total device unlock duration) using UsageStatsManager API	P0
F-48	Track **per-app usage time** for blocked apps (time between successful verification and next lock/app switch)	P1
F-49	Calculate and persist **daily aggregates**: total block attempts, total verified accesses, total screen time, most-attempted app	P0
F-50	Store up to **90 days** of historical usage data locally	P1
F-51	All tracking data stored **locally only** — no cloud upload, no analytics SDK	P0

4.9 Screen-Free Streaks & Positive Reinforcement (V1.5)

ID	Requirement	Priority
F-52	Track **screen-free streaks**: continuous duration where no blocked app was opened	P0
F-53	Show the **current streak** prominently on the parent dashboard (e.g., "2h 15m without blocked apps")	P0
F-54	Send an **encouraging notification** when streak milestones are hit: 1 hour, 2 hours, 4 hours, "all day"	P0
F-55	Buddy character appears in streak notifications with a celebratory message	P1
F-56	Track and display **longest streak ever** and **today's best streak** on the dashboard	P1
F-57	Maintain a **weekly streak summary** (average daily best streak) for trend visibility	P2

4.10 Parent Dashboard & Report Card (V1.5)

ID	Requirement	Priority
F-58	Redesign the dashboard to show a **daily report card** with: blocked app attempts (count), successful verifications, total screen time, current streak, and trend indicators (↑↓ vs. yesterday)	P0
F-59	Show a **"Top Blocked Apps"** section listing the most-attempted blocked apps with attempt counts	P0
F-60	Provide a **weekly summary view** accessible from the dashboard: bar chart or simple visual showing daily block attempts over the past 7 days	P1
F-61	Show Buddy on the dashboard with a **contextual status message**: "Great day! Only 3 attempts" or "Tough day — 15 attempts, but we'll do better tomorrow!"	P1
F-62	**Nudge notifications** for parents: "Your child tried to open TikTok 8 times in the last hour" (configurable threshold)	P1

4.11 Parent Self-Awareness (V1.5)

ID	Requirement	Priority

ID	Requirement	Priority
F-63	**Opt-in feature** (enabled during setup or in settings): track and display the parent's own total daily screen time	P0
F-64	Show parent screen time on the dashboard alongside the child's metrics: "You: 3h 20m · Child: 1h 45m"	P0
F-65	Provide a **gentle, non-judgmental insight** when parent screen time is high: "You've been on for 4+ hours today. Kids learn by watching!" (shown as a Buddy tip, not a nag)	P1
F-66	Allow parent to **set a personal daily screen time goal** (optional) and track progress toward it	P2
F-67	Include parent's screen time in the **weekly summary** for family-wide trend visibility	P1
F-68	Parent self-tracking can be **disabled at any time** from settings with no data loss (data is retained but hidden)	P0

5. Non-Functional Requirements

ID	Requirement	Priority	Version
NF-01	**Performance:** Face recognition inference must complete in < 1.5 seconds on devices with Snapdragon 600-series or equivalent	P0	V1 ☑
NF-02	**Battery:** Background service must consume < 3% battery per hour during idle monitoring	P0	V1 ☑
NF-03	**Storage:** Face embeddings + app data must use < 20 MB total (excluding usage history DB)	P1	V1 ☑
NF-04	**Privacy:** All face data stored locally on-device only; no cloud upload; embeddings stored (not raw images)	P0	V1 ☑
NF-05	**Security:** PIN and face embeddings encrypted at rest (AES-256 via Android Keystore)	P0	V1
NF-06	**Reliability:** Service must survive device reboot, app force-stop attempts, and battery optimization	P0	V1 ☑
NF-07	**Compatibility:** Support Android 8.0 (API 26) through Android 15 (API 35). Note: Android 13+ (API 33) restricts sideloaded apps from granting accessibility service — requires user to manually "Allow restricted settings" from App Info	P0	V1 ☑
NF-08	**Offline:** App must work 100% offline — no internet required (including mascot content and usage tracking)	P0	V1 ☑
NF-08a	**Release build resilience:** R8/ProGuard minification must preserve Gson TypeToken generics, TFLite interpreter classes, ML Kit face detection, CameraX, and Play Core deferred component classes. A comprehensive `proguard-rules.pro` is required. App initialization (`AppState.initialize()`) must be wrapped in try/catch/finally to prevent the app from getting stuck on a loading screen if any platform channel call fails	P0	V1 ☑
NF-09	**Content freshness:** Mascot message library must feel non-repetitive for at least 2 weeks of daily use (minimum 80 unique messages)	P0	V1.5
NF-10	**Usage data storage:** 90 days of usage history must use < 5 MB storage	P1	V1.5
NF-11	**Dashboard load time:** Report card and stats must render in < 1 second	P1	V1.5
NF-12	**Streak accuracy:** Screen-free streak tracking must be accurate within ±30 seconds	P0	V1.5

6. Technical Architecture

6.1 High-Level Architecture

```
┌──────────────────────────────────────────────────────────┐
│                    Flutter UI Layer                      │
│  ┌──────────┐  ┌──────────────┐  ┌────────────────────┐ │
│  │ Onboarding│  │  Dashboard   │  │  Settings Screen   │ │
│  │  Wizard   │  │ + Report Card│  │                    │ │
│  └──────────┘  └──────────────┘  └────────────────────┘ │
│  ┌──────────────────────┐  ┌──────────────────────────┐ │
│  │  Face Registration   │  │   App Selection List     │ │
```

```
│  │    Camera Screen      │  │                          │ │
│  └──────────────────────┘  └──────────────────────────┘ │
│                                                          │
│  ┌──────────────────────┐  ┌──────────────────────────┐ │
│  │  Weekly Summary View │  │  Parent Self-Awareness   │ │
│  │  (Stats + Trends)    │  │  (Screen Time Tracker)   │ │
│  └──────────────────────┘  └──────────────────────────┘ │
├──────────────────────────────────────────────────────────┤
│              Platform Channels (MethodChannel)           │
├──────────────────────────────────────────────────────────┤
│              Native Android Layer (Kotlin)               │
│  ┌──────────────────────────────────────────────────┐   │
│  │         Accessibility Service                    │   │
│  │   (Foreground app detection + event tracking)    │   │
│  └──────────────────────────────────────────────────┘   │
│                                                          │
│  ┌──────────────────────────────────────────────────┐   │
│  │        Foreground Service                        │   │
│  │   (Persistent monitoring + streak tracking)      │   │
│  └──────────────────────────────────────────────────┘   │
│                                                          │
│  ┌──────────────────────────────────────────────────┐   │
│  │       Overlay Activity / Window                  │   │
│  │   (Buddy mascot + face verify + guidance content)│   │
│  └──────────────────────────────────────────────────┘   │
│                                                          │
│  ┌──────────────────────────────────────────────────┐   │
│  │       Face Recognition Engine                    │   │
│  │  ┌──────────────┐    ┌──────────────────────┐    │   │
│  │  │ ML Kit Face  │    │  MobileFaceNet TFLite │    │   │
│  │  │  Detection   │    │  (Embedding Extract)  │    │   │
│  │  └──────────────┘    └──────────────────────┘    │   │
│  └──────────────────────────────────────────────────┘   │
│                                                          │
│  ┌──────────────────────────────────────────────────┐   │
│  │       Usage Tracking Engine                      │   │
│  │   (Block events, screen time, streaks, aggregates)│   │
│  └──────────────────────────────────────────────────┘   │
│                                                          │
│  ┌──────────────────────────────────────────────────┐   │
│  │       Device Admin Receiver                      │   │
│  │   (Uninstall protection + PIN challenge)         │   │
│  └──────────────────────────────────────────────────┘   │
└──────────────────────────────────────────────────────────┘
```

6.2 Key Technical Components

Component	Technology	Purpose	Version
UI Framework	Flutter 3.x + Dart	All user-facing screens	V1 ☑
Platform Bridge	MethodChannel / EventChannel	Flutter ↔ Kotlin communication	V1 ☑
Accessibility Service	Android AccessibilityService (Kotlin)	Detect foreground app changes + log block events	V1 ☑
Background Service	Android Foreground Service (Kotlin)	Keep monitoring alive + streak tracking	V1 ☑
Overlay	System Alert Window / Activity (TYPE_APPLICATION_OVERLAY)	Block screen with Buddy mascot	V1 ☑ / V1.5
Face Detection	Google ML Kit Face Detection	Locate face in camera frame	V1 ☑
Face Recognition	MobileFaceNet (mobilefacenet.tflite) via TFLite Interpreter	Generate 192-d face embedding	V1 ☑
Face Matching	Cosine Similarity (Kotlin)	Compare embeddings (threshold ~0.6–0.7)	V1 ☑
Local Storage	SharedPreferences + Encrypted File	Block list, embeddings, settings	V1 ☑
Auto-restart	BOOT_COMPLETED receiver + WorkManager	Service resilience	V1 ☑
Uninstall Protection	DeviceAdminReceiver + PIN challenge	Prevent app removal	V1 ☑
Mascot Content	Bundled JSON asset	Buddy's message library (80-120 messages)	V1.5
Usage Database	SQLite / Drift (local)	Block events, screen time, streaks, daily aggregates	V1.5
Screen Time Tracking	UsageStatsManager API	Device-wide screen time for parent & child context	V1.5
Streak Engine	Kotlin (in-memory + persisted)	Track continuous screen-free periods	V1.5
Notification Engine	Android NotificationManager	Streak milestones, parent nudges	V1.5

6.3 Face Recognition Pipeline

```
Camera Frame (CameraX)
        │
        ▼
```

```
ML Kit Face Detection
  (locate face bounding box)
        │
        ▼
Crop & Preprocess
  (resize to 112x112, normalize, apply EXIF rotation)
        │
        ▼
MobileFaceNet TFLite Inference
  (output: 192-dimensional embedding)
        │
        ▼
Cosine Similarity vs. stored embeddings
        │
        ▼
  similarity ≥ threshold? ──Yes──▶ GRANT ACCESS
        │                          (log: face_verified)
       No
        ▼
  DENY / Prompt PIN
  (log: face_failed → pin_verified / navigated_away)
```

6.4 App Detection Flow

```
Accessibility Service
  onAccessibilityEvent(TYPE_WINDOW_STATE_CHANGED)
        │
        ▼
Extract package name from event
        │
        ▼
Check against blocked apps list
        │
   Not blocked ──▶ Ignore
        │              │
     Blocked           ▼
        │         Update screen-free
        ▼         streak (no reset)
Check last verification timestamp for this app
        │
   Within interval ──▶ Allow (already verified)
        │
   Expired / Never verified
        │
        ▼
Log block event {app, timestamp}
Reset screen-free streak
        │
        ▼
Launch Blocking Overlay (with Buddy)
```

6.5 Buddy Content Selection Flow (V1.5)

```
Overlay triggered for blocked app
        │
        ▼
Get current time-of-day bucket
  (morning / afternoon / evening / night)
        │
        ▼
Select content category (weighted random):
  ├── Motivational (25%)
  ├── Activity Suggestion (30%) ← time-aware
  ├── Fun Fact (20%)
  ├── Moral Lesson (15%)
  └── Humor (10%)
        │
        ▼
Pick message from category
  (shuffle-bag: no immediate repeats)
        │
        ▼
Select Buddy pose (rotate through 4 poses)
        │
```

```
        ▼
Render overlay:
  ┌──────────────────────────┐
  │   [Buddy illustration]   │
  │                          │
  │   💬 "Did you know       │
  │   dolphins sleep with    │
  │   one eye open?"         │
  │                          │
  │   ── YouTube ──          │
  │   This app is restricted │
  │                          │
  │   [Verify with Face]     │
  │   [Use PIN instead]      │
  └──────────────────────────┘
```

6.6 Usage Tracking Architecture (V1.5)

```
┌──────────────────────────────────────┐
│         Event Sources                │
│  ┌──────────────┐  ┌──────────────────┐  │
│  │Accessibility │  │ UsageStatsManager│  │
│  │  Service     │  │  (Screen Time)   │  │
│  │(block events)│  │                  │  │
│  └──────────────┘  └──────────────────┘  │
│         │                  │             │
│         ▼                  ▼             │
│  ┌──────────────────────────────┐        │
│  │     Usage Tracking Engine    │        │
│  │  ┌────────┐ ┌────────┐ ┌─────┐ │     │
│  │  │ Events │ │Streaks │ │Agg. │ │     │
│  │  │ Logger │ │ Tracker│ │Engine│ │     │
│  │  └────────┘ └────────┘ └─────┘ │     │
│  └──────────────────────────────┘        │
│                │                         │
│                ▼                         │
│  ┌──────────────────────────────┐        │
│  │     Local SQLite Database    │        │
│  │  ├── block_events table      │        │
│  │  ├── daily_aggregates table  │        │
│  │  ├── streaks table           │        │
│  │  └── screen_time table       │        │
│  └──────────────────────────────┘        │
│                │                         │
│                ▼                         │
│  ┌──────────────────────────────┐        │
│  │     Flutter Dashboard        │        │
│  │  (Report Card + Weekly Summary)  │    │
│  └──────────────────────────────┘        │
└──────────────────────────────────────┘
```

7. User Flows

7.1 First-Time Setup Flow

```
Install App
    │
    ▼
Welcome Screen ("Protect your child's screen time")
    │
    ▼
Face Registration (capture parent's face from 3 angles)
    │
    ▼
Add Additional Guardians? (optional, up to 3 more)
    │
    ▼
Set Fallback PIN (minimum 6 digits)
    │
    ▼
Grant Permissions (step-by-step guided flow)
  ├── Accessibility Service
  ├── Display Over Other Apps
```

```
   ├── Device Administrator
   ├── Camera
   ├── Battery Optimization Exemption
   └── Notification
     │
     ▼
Select Apps to Block (scrollable list of installed apps)
     │
     ▼
Setup Complete → Dashboard
     │
     ▼
Background Service Starts
```

7.2 Child Opens Blocked App (V1.5 — with Buddy)

```
Child taps on YouTube (blocked)
     │
     ▼
Accessibility Service detects YouTube in foreground
     │
     ▼
Log block event {youtube, timestamp, pending}
Reset screen-free streak
     │
     ▼
Check: Was YouTube verified in the last [interval]?
     │
    No ──▶ Launch full-screen overlay
     │          │
     │          ▼
     │     Buddy appears with a message:
     │     "How about building something with LEGO instead? 🧱"
     │     ────────────────────
     │     YouTube is restricted
     │     [Verify with Face] [Use PIN]
     │          │
     │     ┌────┴──────────────┐
     │     │                   │
     │   Face Match        No Match
     │     │              (3 attempts)
     │     ▼                   │
     │   Log: face_verified  ▼
     │   Dismiss overlay   Show PIN option
     │   Start timer         │
     │     │            ┌────┴────┐
     │     ▼            │         │
     │   YouTube      PIN OK    PIN Fail
     │   opens          │         │
     │              Log: pin_   Remain
     │              verified   blocked
     │               Dismiss    │
     │               overlay  Log: navigated_away
     │                         (if user goes Home)
     │
    Yes ──▶ Allow through (no overlay)
```

7.3 Parent Views Report Card (V1.5)

```
Parent opens KidShield app
     │
     ▼
Dashboard shows:
   ┌──────────────────────────────────────┐
   │ 🛡 Protection: ON                    │
   │                                      │
   │ TODAY'S REPORT CARD                  │
   │ ├── Block attempts: 12 (↓ vs. yesterday)
   │ ├── Verified access: 4               │
   │ ├── Screen time: 2h 15m             │
   │ ├── Best streak: 3h 40m             │
   │ └── Most attempted: YouTube (7x)    │
   │                                      │
   │ YOUR SCREEN TIME: 3h 20m            │
```

```
    │ "Kids learn by watching! 🧮"        │
    │                                     │
    │ Buddy says: "Good progress today!   │
    │ Only 12 attempts — down from 18!"   │
    │                                     │
    │ [Weekly Summary] [Manage Apps]      │
    │ [Settings]                          │
    └─────────────────────────────────────┘
```

7.4 Parent Accesses Settings

```
Parent opens KidShield app
    │
    ▼
Dashboard shows status + blocked app count
    │
    ▼
Taps "Settings"
    │
    ▼
Face verification or PIN required
    │
    ▼
Settings menu:
  ├── Manage Blocked Apps
  ├── Manage Faces
  ├── Change PIN
  ├── Re-verification Interval
  └── Enable/Disable Protection
```

8. Screen Inventory

#	Screen	Description	Version
S-01	**Welcome / Splash**	App logo, tagline, "Get Started" button	V1 ☑
S-02	**Face Registration**	Camera viewfinder with face alignment guide, capture button, progress indicator (3 captures)	V1 ☑
S-03	**Add More Guardians**	Option to register additional faces or skip	V1 ☑
S-04	**PIN Setup**	6-digit PIN entry with confirmation	V1 ☑
S-05	**Permission Guide**	Step-by-step permission granting with illustrations and "Grant" buttons	V1 ☑
S-06	**App Selection**	List of installed apps with toggle switches, search bar, "Done" button	V1 ☑
S-07	**Dashboard + Report Card**	Protection status, blocked app count, daily report card (attempts, screen time, streaks, trends), parent screen time, Buddy status message	V1 ☑ / V1.5
S-08	**Blocking Overlay (Buddy)**	Full-screen overlay featuring Buddy mascot with speech bubble, app restriction info, "Verify" and "Go Back" buttons	V1 ☑ → V1.5 redesign
S-09	**Face Verification**	Camera viewfinder with real-time face detection feedback, countdown timer	V1 ☑
S-10	**PIN Entry**	6-digit PIN input, "Forgot PIN" info text	V1 ☑
S-11	**Settings**	List of configurable options + parent self-awareness toggle + mascot toggle	V1 ☑ / V1.5
S-12	**Manage Faces**	Grid/list of registered faces with add/delete options	V1 ☑
S-13	**Interval Config**	Text input for seconds + preset chips	V1 ☑
S-14	**Weekly Summary**	7-day view of daily block attempts (bar chart or list), screen time trends, streak records	V1.5

#	Screen	Description	Version
S-15	**Streak Notification**	System notification with Buddy character celebrating streak milestones	V1.5
S-16	**Parent Nudge Notification**	System notification alerting parent of high block-attempt activity	V1.5

9. Data Model

9.1 Local Storage Schema

```
KidShieldData/
├── face_embeddings.enc          # AES-256 encrypted file
│   ├── parent_1: {name, embedding[192], created_at}
│   ├── parent_2: {name, embedding[192], created_at}
│   ├── parent_3: {name, embedding[192], created_at}
│   └── parent_4: {name, embedding[192], created_at}
│
├── preferences (SharedPreferences)
│   ├── pin_hash: String              # SHA-256 hash of PIN
│   ├── is_protection_enabled: Boolean
│   ├── reverification_interval_seconds: Int (default: 1800)
│   ├── is_setup_complete: Boolean
│   ├── debug_mode: Boolean
│   ├── mascot_enabled: Boolean       # V1.5 (default: true)
│   ├── parent_tracking_enabled: Boolean  # V1.5 (default: false, opt-in)
│   ├── parent_screen_time_goal_minutes: Int  # V1.5 (optional)
│   └── nudge_threshold: Int          # V1.5 (default: 5 attempts/hour)
│
├── blocked_apps (SharedPreferences / JSON)
│   └── List<String>             # Package names (e.g., "com.google.youtube")
│
├── verification_sessions (in-memory + SharedPreferences)
│   └── Map<String, Long>        # {package_name: last_verified_timestamp}
│
├── buddy_content.json (bundled asset)        # V1.5
│   └── categories:
│       ├── motivational: [{id, message, time_of_day: "any"}]
│       ├── activity: [{id, message, time_of_day: "morning|afternoon|evening"}]
│       ├── fun_fact: [{id, message, time_of_day: "any"}]
│       ├── moral: [{id, message, time_of_day: "any"}]
│       └── humor: [{id, message, time_of_day: "any"}]
│
├── usage.db (SQLite)                         # V1.5
│   ├── block_events table
│   │   ├── id: INTEGER PRIMARY KEY
│   │   ├── app_package: TEXT
│   │   ├── timestamp: INTEGER (epoch ms)
│   │   └── outcome: TEXT (face_verified / pin_verified / navigated_away)
│   │
│   ├── daily_aggregates table
│   │   ├── date: TEXT (YYYY-MM-DD) PRIMARY KEY
│   │   ├── total_block_attempts: INTEGER
│   │   ├── total_verified: INTEGER
│   │   ├── total_screen_time_minutes: INTEGER
│   │   ├── parent_screen_time_minutes: INTEGER (nullable)
│   │   ├── best_streak_minutes: INTEGER
│   │   └── most_attempted_app: TEXT
│   │
│   ├── streaks table
│   │   ├── id: INTEGER PRIMARY KEY
│   │   ├── start_time: INTEGER (epoch ms)
│   │   ├── end_time: INTEGER (epoch ms, null if active)
│   │   └── duration_minutes: INTEGER
│   │
│   └── screen_time table
│       ├── date: TEXT (YYYY-MM-DD)
│       ├── context: TEXT (child / parent)
│       └── total_minutes: INTEGER
│
└── buddy_state (SharedPreferences)           # V1.5
    ├── last_shown_messages: List<Int>    # IDs of recently shown (prevent repeats)
    ├── current_pose_index: Int
    └── messages_shown_count: Int
```

10. Risk Assessment

Risk	Likelihood	Impact	Mitigation	Version
Android OS kills background service	High	Critical	Foreground service + WorkManager + BOOT_COMPLETED + battery optimization exemption	V1 ☑
Child holds photo of parent to camera	Medium	High	ML Kit liveness detection (blink/head movement) — **V2 feature**	V2
Google Play rejects due to Accessibility Service usage	Medium	High	Clearly document accessibility use; comply with Google's Accessibility API policy; defer Play Store to V2	V2
Face recognition fails in poor lighting	Medium	Medium	PIN fallback always available; guide user to improve lighting	V1 ☑
Device Admin prevents normal uninstall by parent	Low	Medium	PIN challenge on admin disable; clear in-app "Remove Protection" flow	V1 ☑
MobileFaceNet accuracy varies across ethnicities	Medium	Medium	Test broadly; allow cosine similarity threshold tuning in debug mode	V1
Child factory-resets device	Low	High	Out of scope for MVP; document as known limitation	—
Buddy content feels repetitive after extended use	Medium	Medium	Minimum 80 unique messages across 5 categories; shuffle-bag prevents immediate repeats; expand library over time	V1.5
UsageStatsManager permission denied or unavailable	Medium	Low	Graceful degradation — screen time shows "unavailable" but all other features work. Block event tracking uses accessibility service (already granted)	V1.5
Parent finds self-tracking intrusive or naggy	Low	Medium	Strictly opt-in; gentle/humorous tone; easy to disable anytime; no guilt-tripping language	V1.5
Usage database grows too large	Low	Low	Auto-prune data older than 90 days; daily aggregates are compact (~100 bytes/day)	V1.5

11. V1 Rectifications & Lessons Learned

During V1 development and testing, several critical bugs were discovered and resolved. These are documented here for engineering context and to prevent regression.

#	Issue	Root Cause	Fix Applied	Affected Files
R-01	**Face detection silently failing** — front camera images processed by ML Kit returned no faces	`BitmapFactory.decodeByteArray()` does not apply EXIF rotation from JPEG data. Front camera images were sideways/upside-down, making faces undetectable	Added `ExifInterface` parsing + `Matrix` rotation in `PlatformChannelHandler.decodeBitmapWithExifRotation()`. Overlay uses `imageProxy.imageInfo.rotationDegrees` for CameraX frames	PlatformChann BlockingOverla
R-02	**Embedding extraction crash** — TFLite output buffer shape mismatch `[1,192]` vs `[1,128]`	MobileFaceNet model actually outputs 192-dimensional embeddings, not 128 as documented in the original architecture	Changed `EMBEDDING_SIZE` constant from 128 to 192 throughout	FaceRecognitio
R-03	**Battery optimization permission crash** — app crashed when requesting battery optimization exemption	`ACTION_REQUEST_IGNORE_BATTERY_OPTIMIZATIONS` intent was missing `FLAG_ACTIVITY_NEW_TASK` when launched from a non-Activity context (platform channel handler)	Added `Intent.FLAG_ACTIVITY_NEW_TASK` to the battery optimization intent	PlatformChann

#	Issue	Root Cause	Fix Applied	Affected Files
R-04	**Onboarding always restarts** — every app launch showed face registration even if already completed	Welcome screen was a stateless widget that always navigated to step 1. No check for existing faces or PIN	Converted to StatefulWidget. `initState()` checks existing faces/PIN via platform channel and skips to the first incomplete onboarding step	welcome_scree
R-05	**Verification loop** — after successful face verify, overlay immediately re-appeared for the same app	On overlay dismiss, `resetLastDetected()` cleared the tracked package, causing the accessibility service to re-detect the same blocked app in foreground and re-trigger the overlay	Replaced `resetLastDetected()` with `overlayActive` flag + `onOverlayDismissed()` callback pattern. Added 3-second grace period for "every time" mode. Added `Handler`-based recheck timer instead of immediate re-detection	KidShieldAcces BlockingOverla
R-06	**UI overflow on PIN screen** — keyboard pushed content off-screen	PIN entry screen used a fixed layout without scroll wrapping	Wrapped in `SafeArea` + `SingleChildScrollView`	pin_setup_scree
R-07	**UI overflow on guardians screen** — buttons pushed off-screen with long face list	Layout used `Spacer()` between content and buttons. On small screens with multiple registered faces, this caused RenderFlex overflow	Replaced `Spacer()` with scrollable content area + pinned bottom buttons using `Column` + `Expanded` + `SingleChildScrollView`	add_guardians_
R-08	**Release build stuck on loading** — app showed infinite loading spinner in release mode	R8 minification stripped Gson `TypeToken` generics, TFLite interpreter classes, and ML Kit classes. Platform channel calls threw exceptions silently. `AppState.initialize()` had no error handling, so `isLoading` stayed `true` forever	Created comprehensive `proguard-rules.pro` with keep rules for Gson, TFLite, ML Kit, CameraX, Flutter, Play Core. Wrapped `AppState.initialize()` in try/catch/finally so loading completes even on failure	proguard-rules build.gradle.kts
R-09	**Accessibility service ungrantable on Android 13+** — sideloaded APK could not enable accessibility service	Android 13 (API 33) introduced "restricted settings" that blocks sideloaded apps from granting accessibility/device admin permissions	Added instructional text on the permission screen guiding users to App Info → "Allow restricted settings." This is an OS-level limitation with no programmatic workaround	permission_scr

#	Issue	Root Cause	Fix Applied	Affected Files
R-10	**No way to skip onboarding steps** — user forced through all steps even if partially complete or wanting to skip	All onboarding screens required completion to proceed	Added skip buttons to face registration (if faces exist) and permission screen (with confirmation dialog warning about reduced protection)	face_registratio permission_scr

12. Scope Boundaries

Completed (V1 / MVP) ☑

- Face registration (multiple adults, MobileFaceNet 192-d embeddings) + PIN fallback
- Permission acquisition wizard (camera, accessibility, overlay, device admin, battery, notifications)
- Manual app block-list selection with search
- Background accessibility-based monitoring (foreground service + auto-restart)
- Full-screen blocking overlay with face verification + PIN fallback
- Configurable re-verification interval (seconds-based, with presets)
- Device admin uninstall protection with PIN challenge
- 100% offline operation
- Debug mode (7-tap)
- ProGuard/R8 release build configuration

In Scope (V1.5 — Current Phase)

- **Buddy mascot character** on blocking overlay (static illustration + rotating text bubbles)
- **Content library** (80-120 messages: motivational, activities, fun facts, moral lessons, humor)
- **Time-of-day-aware** content selection
- **Usage tracking** (block events, outcomes, per-app stats)
- **Screen time tracking** (device-wide via UsageStatsManager)
- **Screen-free streak** tracking + milestone notifications
- **Dashboard report card** (daily stats, trends, most-attempted apps)
- **Weekly summary view** (7-day trends)
- **Parent self-awareness** (opt-in screen time tracking + gentle insights)
- **Parent nudge notifications** (configurable threshold for high activity alerts)

Out of Scope (Future Versions)

Feature	Target Version
Liveness detection (anti-spoofing)	V2
Animated Buddy mascot (Lottie/Rive)	V2
AI-powered contextual Buddy responses	V2
Voice lines / TTS for Buddy	V2
Scheduled free-time windows	V2
Per-app time limits	V2
"Earn screen time" via offline challenges	V2
Family challenges ("Tech-free dinner streak")	V2
New app install notifications	V2
Parent screen time goal tracking	V2
Detailed per-app analytics with charts	V2
Google Play Store distribution	V2
iOS support	V3+
Remote management (parent's phone → child's phone)	V3+
Multiple child profiles	V3+
Cloud sync / backup	V3+

13. Development Phases

Phase 1 — Foundation (Week 1–2) ☑ COMPLETE

- [x] Flutter project setup with Kotlin native module
- [x] Platform channel architecture (MethodChannel / EventChannel)
- [x] Permission request flow (all required permissions)
- [x] Local storage layer (SharedPreferences, face embeddings)

Phase 2 — Face Recognition (Week 3–4) ☑ COMPLETE

- [x] CameraX integration for face capture
- [x] ML Kit face detection integration
- [x] MobileFaceNet TFLite integration (192-d embedding extraction)
- [x] EXIF rotation handling for front camera
- [x] Face registration flow (capture, process, store)
- [x] Face verification flow (capture, compare, threshold)
- [x] PIN setup and verification

Phase 3 — Core Blocking (Week 5–6) ☑ COMPLETE

- [x] Accessibility Service implementation (foreground app detection)
- [x] Foreground Service (persistent background monitoring)
- [x] Blocking overlay (system alert window, undismissable)
- [x] App selection screen (list installed apps, toggle block, search)
- [x] Verification session management (interval-based re-prompting, seconds-based)

Phase 4 — Resilience & Settings (Week 7–8) ☑ COMPLETE

- [x] Device Admin Receiver (uninstall protection + PIN challenge)
- [x] Auto-restart mechanisms (BOOT_COMPLETED + START_STICKY)
- [x] Settings screen (interval config, manage faces, manage apps)
- [x] Dashboard screen with protection status
- [x] Debug/testing mode (7-tap)
- [x] ProGuard/R8 configuration for release builds

Phase 5 — Polish & Testing (Week 9–10) ☑ COMPLETE

- [x] End-to-end testing (face detection, overlay, verification loop)
- [x] Edge case handling (permissions, camera failure, EXIF rotation)
- [x] UI polish (scrollable layouts, keyboard overflow fixes, skip options)
- [x] Release APK build and sideload testing
- [x] Restricted settings guidance for Android 13+

Phase 6 — Buddy Mascot & Overlay Redesign (Week 11–12)

- [ ] Design Buddy character illustrations (human child, 4 poses: waving, thinking, pointing, encouraging)
- [ ] Create and curate content library JSON (80-120 messages across 5 categories)
- [ ] Implement time-of-day content bucketing (morning/afternoon/evening/night)
- [ ] Implement shuffle-bag message selection (no immediate repeats)
- [ ] Redesign blocking overlay layout: Buddy illustration + speech bubble + app info + verification controls
- [ ] Add mascot enable/disable toggle in settings
- [ ] Test overlay with various screen sizes and orientations

Phase 7 — Usage Tracking & Streak Engine (Week 13–14)

- [ ] Set up local SQLite database (Drift or sqflite) with schema: block_events, daily_aggregates, streaks, screen_time
- [ ] Instrument accessibility service to log block events with outcomes
- [ ] Implement screen-free streak tracking (start/stop/reset logic in foreground service)
- [ ] Integrate UsageStatsManager for device-wide screen time (with graceful fallback)
- [ ] Build daily aggregation engine (runs at midnight or on app open)
- [ ] Implement 90-day data retention with auto-pruning
- [ ] Add platform channel methods for Flutter to query usage data

Phase 8 — Dashboard Report Card & Notifications (Week 15–16)

- [ ] Redesign dashboard with daily report card (attempts, verified, screen time, streak, trends)
- [ ] Build "Top Blocked Apps" section with attempt counts
- [ ] Add trend indicators (↑↓ vs. yesterday for each metric)
- [ ] Build weekly summary view (7-day bar chart or list view)
- [ ] Add Buddy contextual status message on dashboard
- [ ] Implement streak milestone notifications (1h, 2h, 4h, all-day) with Buddy character

- ☐ Implement parent nudge notifications (configurable threshold)

Phase 9 — Parent Self-Awareness (Week 17–18)

- ☐ Add opt-in parent screen time tracking toggle (setup + settings)
- ☐ Integrate UsageStatsManager for parent context (track when app is in foreground after face/PIN verify)
- ☐ Display parent screen time on dashboard alongside child metrics
- ☐ Implement gentle insight messages for high parent screen time
- ☐ Include parent data in weekly summary
- ☐ Add disable option in settings (retain data, just hide display)

Phase 10 — V1.5 Polish & Release (Week 19–20)

- ☐ End-to-end testing of all V1.5 features
- ☐ Performance profiling (database queries, dashboard render time)
- ☐ Battery impact assessment with new tracking features
- ☐ Content review: ensure all 80-120 Buddy messages are appropriate, diverse, and engaging
- ☐ UI/UX polish across new screens
- ☐ Release APK build and testing

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14. Glossary

Term	Definition
Accessibility Service	Android API that allows an app to observe UI events like window changes across the system
Buddy	KidShield's mascot character — a relatable human child illustration that appears on the blocking overlay, providing encouragement, fun facts, activity suggestions, and moral lessons
Content Library	A bundled JSON asset containing 80-120 pre-written messages across 5 categories (motivational, activities, fun facts, moral lessons, humor) that Buddy displays on the overlay
Cosine Similarity	A metric measuring the angle between two vectors; used to compare face embeddings (1.0 = identical, 0.0 = unrelated)
Daily Aggregate	A summarized row of usage data per day: total block attempts, verified accesses, screen time, best streak, most-attempted app
Device Admin	Android API that grants an app elevated privileges, including preventing its own uninstall
Foreground Service	An Android service that shows a persistent notification and is less likely to be killed by the OS
MobileFaceNet	A lightweight face recognition neural network optimized for mobile devices; outputs a 192-d embedding per face
Overlay / SYSTEM_ALERT_WINDOW	Permission allowing an app to draw on top of other apps
Parent Self-Awareness	An opt-in feature that tracks and surfaces the parent's own daily screen time alongside the child's metrics, encouraging family-wide digital wellness
Platform Channel	Flutter mechanism for calling native Android/iOS code from Dart
Screen-Free Streak	A continuous period during which no blocked app was opened; tracked and celebrated with milestone notifications
Shuffle-Bag Algorithm	A randomization technique that ensures all items in a set are shown once before any repeats — used for Buddy's message rotation to maximize variety
Time-of-Day Bucketing	Categorizing content selection based on the current time: morning (6am–12pm), afternoon (12pm–5pm), evening (5pm–9pm), night (9pm–6am) — used to make Buddy's activity suggestions contextually appropriate
UsageStatsManager	Android API that provides access to device usage history data, including app usage durations and device unlock events

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This document was last updated on March 3, 2026. V1 (MVP) is complete. V1.5 development is in progress.